(ii) Electric field due to a long charged cylinder/infinite rod of charge

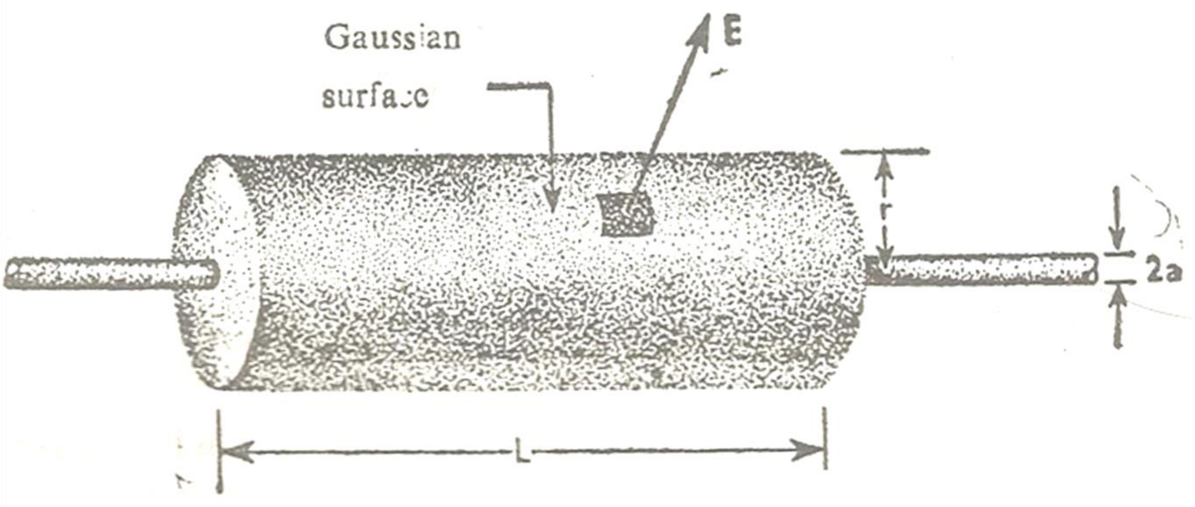

Fig. 1. Cylindrical Gaussian surface for a long charged cylinder.

Fig. 1. shows a long cylinder/a section of an infinite rod of charge of radius *a*, uniformly charged having a charge λ per unit length. The field $\vec{E}$ at any point outside the cylinder can be obtained by constructing a Gaussian surface that passes through that point and surrounds an arbitrary length *L* of the cylinder.

The field is everywhere constant on the Gaussian surface, and directed radially away from the axis so that $\vec{E}$ and $d\vec{a}$ are in the same direction on the curved surface. Hence, from Gauss law

$$\oint_S \vec{E}.d\vec{a} = \oint_S E da = E\oint_a da = \frac{\lambda}{\varepsilon_0}L$$

$$E(2\pi rL) = \frac{\lambda}{\varepsilon_0}L$$

Where the integration is taken over the curved surface. Thus

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

In vector notation

$$\vec{E} = \frac{\lambda}{2\pi\varepsilon_0}\frac{\vec{r}}{r^2}$$

Flux of a cylindrical Gaussian surface

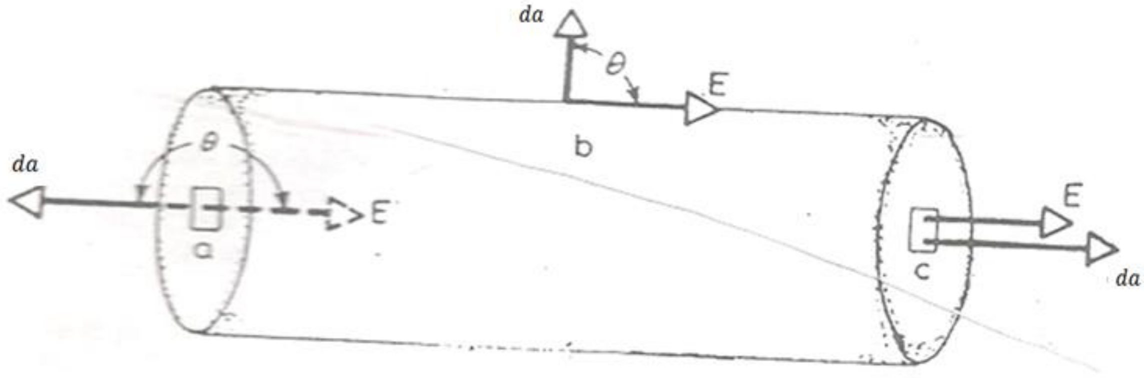

Fig. 2.

Fig. 2. shows a hypothetical cylinder of radius R immersed in a uniform electric field $\vec{E}$. The cylinder axis is parallel to the field. As shown from the figure there are three surfaces

(i) the left cap *a*
(ii) the cylindrical surface *b*
(iii) the right cap *c*.

Therefore $$\varphi_E = \oint_S \vec{E}.d\vec{a}$$

$$= \int_a \vec{E}.d\vec{a} + \int_b \vec{E}.d\vec{a} + \int_c \vec{E}.d\vec{a}$$

For the left cap, angle θ between $\vec{E}$ and $d\vec{a}$, is 180^0 for all points, *i.e.*, $\vec{E}$ and $d\vec{a}$ are oppositely directed. Thus

$$\int_a \vec{E}.d\vec{a} = \int_a E\cos 180^0 da = -E\int_a da = -Ea$$

Where $a = \pi R^2$ is the cap area. Similarly, for right cap,

$$\int_c \vec{E}.d\vec{a} = Ea$$

Since the angle $\theta = 0$. Finally for cylindrical wall $\theta = 90^0$. Hence we get,

$$\int_b \vec{E}.d\vec{a} = \int_b E\cos 90^0 da = 0$$

Thus, the total flux

$$\varphi_E = -Ea + 0 + Ea = 0$$

Thus the net flux of a cylindrical Gaussian surface placed in an external electric field is zero.

Dielectric material: The material through which electricity cannot pass is called the dielectric material. In this material, the electrons are combined in atoms at a certain distance, *i.e.*, electrons are not free to move from one atom to another. For this reason, they cannot carry electricity. Rubber, amber, glasses are known as dielectric materials.

Dielectric constant: The ratio of the capacitance C with the dielectric to that without C_0 is called the dielectric constant k, which is the property of the dielectric material *i.e.*,

$$k = \frac{C}{C_0}$$

k is called the *relative permittivity* of the medium.

Gauss's law for dielectric

Let us apply Gauss's law for parallel plate capacitor filled with a dielectric of dielectric constant k. Fig. 3. (a) shows a parallel plate capacitor without dielectric. Let charge is q on the plates with proper sign and A is the area of each plate. E_0 is the electric field without dielectric.

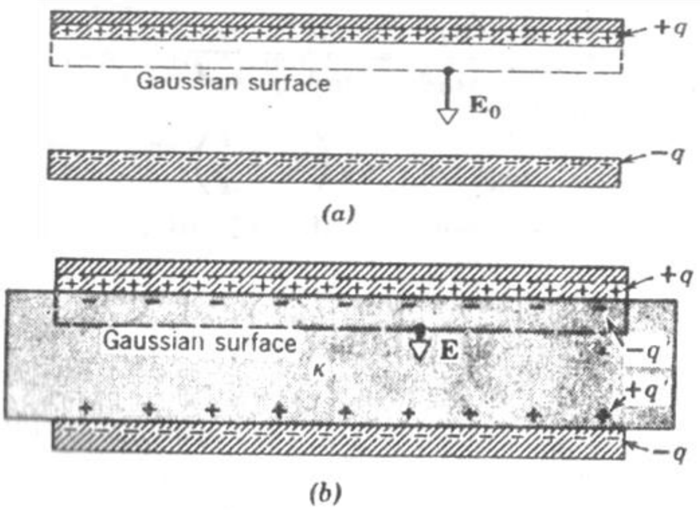

Fig. 3. A parallel plate capacitor (a) without dielectric (b) with dielectric. The charge q on the plates is assumed to be the same in each case.

Therefore from the Gauss's law

$$\varepsilon_0 \oint \vec{E}.d\vec{a} = \varepsilon_0 E_0 A = q$$

$$\therefore \qquad E_0 = \frac{q}{\varepsilon_0 A}$$

Now place the dielectric as shown in fig. 3. (b) in which $-q^{/}$ is the *induced surface charge,* must be distinguished from q. The two charges, both of which lie within the Gaussian surface, are of opposite sign. Then the net charge enclosed by Gaussian surface is $q - q^{/}$. Therefore with dielectric the Gauss's law is,

$$\varepsilon_0 \oint \vec{E}.d\vec{a} = q - q^{/} \qquad (1)$$

$$Or \qquad \varepsilon_0 EA = q - q^{/}$$

$$\therefore \qquad E = \frac{q}{\varepsilon_0 A} - \frac{q^/}{\varepsilon_0 A} \qquad (2)$$

In which $-q^/$ is the induced charge. We can write

$$E = \frac{E_0}{k}$$

Where, E is the field inside the dielectric substituting E_0, we have

$$E = \frac{q}{k\varepsilon_0 A} \qquad (3)$$

Substituting this value of E in equation (2) we get

$$\frac{q}{k\varepsilon_0 A} = \frac{q}{\varepsilon_0 A} - \frac{q^/}{\varepsilon_0 A} \qquad (4)$$

$$\therefore \qquad q^/ = q\left(1 - \frac{1}{k}\right) \qquad (5)$$

If there is no dielectric between the plates then $k = 1$, and $q^/ = 0$, *i.e.*, there will be no induced charge. The value of k is always greater than one $(k > 1)$, hence $q^/$ *(induced charge)* is always less than *q (free charge)*.

From equation (5) the value of $q - q^/$ is given by

$$q - q^/ = \frac{q}{k}$$

Now we can write the Gauss's law with dielectric as given in equation (1) after substituting $q - q^/$in the following form.

$$\varepsilon_0 \oint \vec{E}.d\vec{a} = \frac{q}{k}$$

$$\therefore \qquad \varepsilon_0 \oint k\vec{E}.d\vec{a} = q \qquad (6)$$